PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

25 Jan – 31 Jan 2027 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
25 Jan D2027-01-25	Daily mixed Daily test	___ / 10	Retest: 28 Jan
27 Jan D2027-01-27	Daily mixed Daily test	___ / 10	Retest: 30 Jan
28 Jan D2027-01-28	Daily mixed Daily test	___ / 10	Retest: 31 Jan
29 Jan D2027-01-29	Daily mixed Daily test	___ / 10	Retest: 1 Feb
30 Jan D2027-01-30	Daily mixed Daily test	___ / 10	Retest: 2 Feb
31 Jan T2027-01-31	Science Full paper (3 hr)	___ / 80	Retest: 3 Feb

Daily Test · Answer key

Monday, 25 January 2027 · 10 marks

Test ID: D2027-01-25

Maths 5.1–5.2 Circle basics · AI Self-Management · English Unit 3

Q1 [1]

Question: A circle has rotational symmetry about its _____ through any angle.

Expected answer / marking points: centre

Q2 [1]

Question: How many lines of reflection symmetry does a circle have?

Expected answer / marking points: Infinitely many — every diameter is a line of symmetry.

Q3 [1]

Question: How many circles can pass through two given points? Where do their centres lie?

Expected answer / marking points: Infinitely many; centres lie on the perpendicular bisector of the segment joining the two points.

Q4 [2]

Question: Write two signs of stress in a student and two healthy ways to manage it.

Expected answer / marking points: Signs: headache, poor sleep, irritability, unable to concentrate (any two) [1]; ways: exercise, enough sleep, time table, talk to someone, breaks/hobbies (any two) [1].

Q5 [1]

Question: What is self-awareness?

Expected answer / marking points: Knowing your own strengths, weaknesses, feelings and interests.

Q6 [1]

Question: Write one sign of a self-confident person.

Expected answer / marking points: Any one: makes eye contact, speaks clearly, accepts mistakes, tries new things.

Q7 [1]

Question: In “Winds of Change”, what is a *pankha*?

Expected answer / marking points: A traditional hand fan.

Q8 [1]

Question: Why did pankhas from different regions of India look different from one another?

Expected answer / marking points: Each place used local materials (sola, bamboo, beads, leather) and its own designs.

Q9 [1]

Question: In the poem “Canvas of Soil”, what is compared to brushstrokes in “brushstrokes of seeds”?

Expected answer / marking points: The seeds sown by the gardener (metaphor).

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Wednesday, 27 January 2027 · 10 marks

Test ID: D2027-01-27

Hindi पाठ 5 · SST Ch 5 State & Society · Science Ch 5 Mixtures

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Q1 [1]

Question: इस पाठ के लेखक कौन हैं?

Expected answer / marking points: मोहन राकेश।

Q2 [2]

Question: कन्याकुमारी को 'आखिरी चट्टान' क्यों कहा गया है?

Expected answer / marking points: यह भारत की मुख्य भूमि का सबसे दक्षिणी छोर है [1]; यहाँ तीन समुद्रों का संगम है और आगे केवल समुद्र है [1]।

Q3 [3]

Question: How did trade and guilds help the economy of early India?

Expected answer / marking points: Guilds (śreṇīs) organised crafts and set standards [1]; trade routes and ports linked India with other lands [1]; this brought wealth, crafts and cultural exchange [1].

Q4 [2]

Question: How will you separate a mixture of salt and sand?

Expected answer / marking points: Add water and stir — salt dissolves [0.5]; filter to remove sand [0.5]; evaporate the filtrate to get salt [1].

Q5 [2]

Question: Find the concentration (mass by mass %) of a solution with 20 g of salt in 180 g of water.

Expected answer / marking points: Solution = 200 g [1]; 20/200 × 100 = 10% [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Thursday, 28 January 2027 · 10 marks

Test ID: D2027-01-28

SST Ch 6 Democracy · Science Ch 6 Forces · Maths Ch 6 Perimeter & Area

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Q1 **[3]** **Question:** How did ancient Indian traditions contain democratic ideas? Give three points.

Expected answer / marking points: Sabhā and samiti allowed consultation [1]; gaṇas/saṁghas were ruled by assemblies [1]; collective decision-making and accountability of the rājā [1].

Q2 **[2]** **Question:** Two forces of 8 N and 5 N act on a body in opposite directions. Find the net force and its direction. Is the body in equilibrium?

Expected answer / marking points: Net = 3 N in the direction of the 8 N force [1]; no, forces are unbalanced [1].

Q3 **[2]** **Question:** Why is it easier to push an empty trolley than a loaded one?

Expected answer / marking points: Loaded trolley has more mass, so more inertia [1]; for the same acceleration it needs more force (F = ma) [1].

Q4 **[3]** **Question:** A circular park of radius 21 m has a 3.5 m wide path outside it. Find the area of the path.

Expected answer / marking points: Outer radius 24.5 m [1]; area = $22/7 \times (24.5^2 - 21^2) = 22/7 \times 45.5 \times 3.5$ [1] = 500.5 m^2 [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Friday, 29 January 2027 · 10 marks

Test ID: D2027-01-29

Maths Ch 7 Probability · AI Entrepreneurial Skills · English Unit 5

Q1 **[3]**

Question: Two dice are thrown. Find the probability that the sum is (i) 7 (ii) 12 (iii) less than 4.

Expected answer / marking points: 36 outcomes. (i) 6/36 = 1/6 [1] (ii) 1/36 [1] (iii) sums 2, 3: (1,1), (1,2), (2,1) ⇒ 3/36 = 1/12 [1].

Q2 **[3]**

Question: A box has 3 red and 2 blue pens. Two pens are taken one after another without replacement. Using a tree diagram, find P(both blue) and P(one red and one blue).

Expected answer / marking points: P(BB) = 2/5 × 1/4 = 1/10 [1.5]; P(RB or BR) = 3/5 × 2/4 + 2/5 × 3/4 = 12/20 = 3/5 [1.5].

Q3 **[2]**

Question: How do entrepreneurs help society?

Expected answer / marking points: Create jobs [1]; provide new products/services and solve problems [1].

Q4 **[2]**

Question: How do the Paralympics change people's attitudes?

Expected answer / marking points: People see the strength and skill of para-athletes [1]; this breaks stereotypes about disability and promotes inclusivity [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Saturday, 30 January 2027 · 10 marks

Test ID: D2027-01-30

Hindi पाठ 7 मैं और मेरा देश · SST Ch 7 Elections · Science Ch 7 Work & Energy

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Q1 **Question:** लेखक ने 'शक्तिबोध' और 'सौंदर्यबोध' से क्या समझाया है?
[2]

Expected answer / marking points: शक्तिबोध — देश की ताकत/आत्मविश्वास को बढ़ाने वाले काम [1]; सौंदर्यबोध — देश की छवि, स्वच्छता और सुंदरता बनाए रखना [1]।

Q2 **Question:** लेखक को अपनी स्थिति 'हीन' क्यों लगी?
[2]

Expected answer / marking points: देश पराधीन था [1]; इसलिए विदेश में या कहीं भी कोई भी उसका अपमान कर सकता था और वह कुछ नहीं कर सकता था [1]।

Q3 **Question:** What is the Model Code of Conduct?
[2]

Expected answer / marking points: A set of rules for parties and candidates during elections [1], e.g., no use of government resources for campaigning, no appeal on caste/religion [1].

Q4 **Question:** Give two differences between national and state parties.
[2]

Expected answer / marking points: National parties are recognised in several states and contest across India; state parties mainly in one state [1]; national parties focus on all-India issues, state parties on regional issues [1].

Q5 **Question:** Find the kinetic energy of a 50 kg boy running at 4 m s^{-1}.
[2]

Expected answer / marking points: ½ × 50 × 16 [1] = 400 J [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Science · Full Paper 1 (Annual pattern) · Answer key

Sunday, 31 January 2027 · 80 marks

Test ID: T2027-01-31

Class 9 · Exploration — poora syllabus (Ch 1–13)

Q1 [1]

Question: Which step of the scientific method comes immediately after making a hypothesis?

(a) Observation
(b) Testing by experiment
(c) Conclusion
(d) Asking a question

Expected answer / marking points: (b) Testing by experiment

Q2 [1]

Question: Ribosomes are the site of:

(a) respiration
(b) photosynthesis
(c) protein synthesis
(d) storage of food

Expected answer / marking points: (c) protein synthesis

Q3 [1]

Question: The tissue that joins muscle to bone is:

(a) ligament
(b) tendon
(c) cartilage
(d) areolar tissue

Expected answer / marking points: (b) tendon

Q4 [1]

Question: The area under a velocity–time graph gives:

(a) acceleration
(b) speed
(c) displacement
(d) force

Expected answer / marking points: (c) displacement

Q5 [1]

Question: Which of the following shows the Tyndall effect?

(a) Salt solution
(b) Sugar solution
(c) Milk
(d) Copper sulfate solution

Expected answer / marking points: (c) Milk

Q6 [1]

Question: A 5 kg body has an acceleration of 2 m s^{-2}. The net force on it is:

(a) 2.5 N
(b) 7 N
(c) 10 N
(d) 3 N

Expected answer / marking points: (c) 10 N

Q7 [1]

Question: The kinetic energy of a 2 kg ball moving at 3 m s^{-1} is:

(a) 6 J
(b) 9 J
(c) 18 J
(d) 3 J

Expected answer / marking points: (b) 9 J

Q8 **Question:** The isotopes of an element have the same:
[1]

(a) mass number
(b) number of neutrons
(c) atomic number
(d) nuclear mass

Expected answer / marking points: (c) atomic number

Q9 **Question:** The valency of nitrogen in NH_3 is:
[1]

(a) 1
(b) 2
(c) 3
(d) 5

Expected answer / marking points: (c) 3

Q10 **Question:** The formula unit mass of NaCl (Na = 23, Cl = 35.5) is:
[1]

(a) 58.5 u
(b) 23 u
(c) 35.5 u
(d) 46 u

Expected answer / marking points: (a) 58.5 u

Q11 **Question:** Humans can hear sounds of frequency:
[1]

(a) below 20 Hz
(b) 20 Hz to 20 kHz
(c) above 20 kHz
(d) only above 1 kHz

Expected answer / marking points: (b) 20 Hz to 20 kHz

Q12 **Question:** Persistence of sound in a hall due to repeated reflections is called:
[1]

(a) echo
(b) reverberation
(c) resonance
(d) refraction

Expected answer / marking points: (b) reverberation

Q13 **Question:** In flowering plants, the ovule develops into the:
[1]

(a) fruit
(b) seed
(c) flower
(d) pollen

Expected answer / marking points: (b) seed

Q14 **Question:** Organisms with prokaryotic cells are placed in kingdom:
[1]

(a) Protista
(b) Fungi
(c) Monera
(d) Plantae

Expected answer / marking points: (c) Monera

Q15 **Question:** Fraction of sunlight reflected by a surface is its:
[1]

(a) albedo
(b) humidity
(c) latitude
(d) insolation

Expected answer / marking points: (a) albedo

Q16 **Question:** Which process removes carbon dioxide from the atmosphere?
[1]

(a) Respiration (b) Burning of fuels

(c) Photosynthesis (d) Decomposition

Expected answer / marking points: (c) Photosynthesis

Q17 **Question: A:** A person sitting in a moving bus falls backward when the bus starts suddenly.
[1] **R:** The lower part of the body moves with the bus, but the upper part tends to remain at rest due to inertia.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q18 **Question: A:** Work done by gravity on a satellite moving in a circular orbit is zero.
[1] **R:** Gravitational force on the satellite is perpendicular to its displacement at every point.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q19 **Question: A:** Sound travels faster in air than in steel.
[1] **R:** Particles are closer and interact more strongly in solids.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (d) — sound is faster in steel; R is true.

Q20 **Question: A:** Burning large amounts of fossil fuels adds to global warming.
[1] **R:** Carbon dioxide is a greenhouse gas that traps heat re-radiated by the Earth.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q21 **Question:** Write two differences between plant and animal tissues.
[2]

Expected answer / marking points: Plant tissues: much dead supportive tissue, growth limited to meristems [1]; animal tissues: mostly living cells, growth uniform; no meristem-like regions (any two correct) [1].

Q22 [2] **Question:** A car travels 100 m north and then 100 m back south in 40 s. Find its average speed and average velocity.

Expected answer / marking points: Average speed = 200/40 = 5 m s^{-1} [1]; displacement 0 ⇒ average velocity 0 [1].

Q23 [2] **Question:** How will you separate (i) iron filings from sand (ii) cream from milk?

Expected answer / marking points: (i) Magnet [1] (ii) Centrifugation [1].

Q24 [2] **Question:** An atom has mass number 27 and 14 neutrons. Find its atomic number and electronic configuration.

Expected answer / marking points: Z = 27 – 14 = 13 [1]; 2, 8, 3 [1].

Q25 [2] **Question:** Give two reasons why biodiversity is important for an ecosystem.

Expected answer / marking points: Maintains food chains/ecological balance [1]; provides resources (food, medicine) and stability against change [1].

Q26 [2] **Question:** How is a land breeze formed at night?

Expected answer / marking points: At night land cools faster than sea [1]; air over the warmer sea rises and cooler air from land blows towards the sea [1].

Q27 [3] **Question:** Draw a labelled diagram of an animal cell showing the nucleus, mitochondria and cell membrane. Write one function of each.

Expected answer / marking points: Diagram [1.5]; nucleus — controls cell activities/has genetic material; mitochondria — releases energy (respiration); cell membrane — controls entry/exit of substances [1.5].

Q28 [3] **Question:** A bus starting from rest moves with uniform acceleration of 0.5 m s^{-2} for 1 minute. Find (i) the speed acquired (ii) the distance travelled.

Expected answer / marking points: t = 60 s [0.5]; (i) v = 0.5 × 60 = 30 m s^{-1} [1]; (ii) s = ½ × 0.5 × 3600 = 900 m [1.5].

Q29 [3] **Question:** A solution has 40 g of common salt in 320 g of water. Find the mass by mass percentage. Classify as true solution, suspension or colloid: blood, brass, chalk in water.

Expected answer / marking points: 40/360 × 100 ≈ 11.1% [1.5]; brass — solution; chalk in water — suspension; blood — colloid (heterogeneous) [1.5].

Q30 [3] **Question:** A bullet of mass 20 g is fired from a gun of mass 4 kg with a velocity of 400 m s^{-1}. Find the recoil velocity of the gun.

Expected answer / marking points: Initial momentum 0 [0.5]; 0.02 × 400 + 4 × v = 0 [1.5] ⇒ v = –2 m s^{-1} (2 m s^{-1} backward) [1].

Q31 [3] **Question:** Calculate the molecular masses of C_2H_5OH and $CaCO_3$. (C = 12, H = 1, O = 16, Ca = 40)

Expected answer / marking points: C_2H_5OH = 24 + 6 + 16 = 46 u [1.5]; $CaCO_3$ = 40 + 12 + 48 = 100 u [1.5].

Q32 [3] **Question:** A person standing 170 m from a cliff claps. After how much time will he hear the echo (speed of sound 340 m s^{-1})? Write two uses of ultrasound.

Expected answer / marking points: t = 2d/v = 340/340 = 1 s [1.5]; ultrasound scans, cleaning parts, detecting cracks in metals, SONAR (any two) [1.5].

Q33 [3] **Question:** Describe binary fission in Amoeba. How is it different from multiple fission?

Expected answer / marking points: Nucleus divides into two, then cytoplasm divides; two daughter cells form [2]; in multiple fission the nucleus divides many times and many daughter cells form at once (e.g., Plasmodium) [1].

Q34 [5] **Question:** State the three laws of motion with one example each. Show that the second law gives F = ma.

ATHVA / OR

Define work, energy and power with their SI units. A 50 kg girl climbs 4 m in 8 s. Find the work done and power. Explain how a pulley and an inclined plane make work easier.

Expected answer / marking points: First law + example (passengers jerk forward) [1]; second law + example (catching a ball moving hands back) [1]; third law + example (rocket, gun recoil) [1]; derivation $F \propto (mv - mu)/t = ma$, k = 1 [2].
OR: Definitions with units [1.5]; W = 50 × 10 × 4 = 2000 J, P = 250 W [1.5]; pulley changes direction of force/reduces effort; inclined plane needs less force over longer distance [2].

Q35 [5] **Question:** Explain how ionic and covalent bonds are formed, with one example each (electron-dot diagrams). Write two properties of each type of compound.

Expected answer / marking points: Ionic: transfer of electrons, e.g., NaCl — $Na \rightarrow Na^+$, $Cl \rightarrow Cl^-$ (diagram) [1.5]; covalent: sharing, e.g., H_2O/CH_4/Cl_2 (diagram) [1.5]; ionic: high m.p., conduct when molten/dissolved; covalent: low m.p., poor conductors [2].

Q36 [5] **Question:** Describe the five-kingdom classification. Give two features and one example of each kingdom.

ATHVA / OR

Describe the carbon cycle and the nitrogen cycle. How do human activities disturb the carbon cycle?

Expected answer / marking points: Monera — prokaryotic, unicellular (bacteria) [1]; Protista — unicellular eukaryotes (Amoeba) [1]; Fungi — heterotrophic, cell wall of chitin (mushroom) [1]; Plantae — multicellular autotrophs, cellulose cell wall (mango) [1]; Animalia — multicellular heterotrophs, no cell wall (dog) [1].
OR: Carbon cycle: CO_2 taken in by photosynthesis, returned by respiration, decomposition, burning; stored in oceans/fossil fuels [2]; nitrogen cycle: fixation by bacteria/lightning, nitrates absorbed by plants, decomposition, denitrification [2]; human impact: burning fossil fuels and deforestation raise CO_2 → global warming [1].

Q37 **Question:**

[4]

> A ball is dropped from the top of a 45 m tall building. Ignore air resistance; g = 10 m s^{-2}.

(i) How long does it take to reach the ground? [1]
(ii) With what speed does it hit the ground? [1]
(iii) Sketch its velocity–time graph and state what the slope of the graph represents. What is the distance covered in the first 2 s? [2]

Expected answer / marking points: (i) $45 = ½ \times 10 \times t^2 \Rightarrow t = 3$ s [1]. (ii) $v = 30$ m s^{-1} [1]. (iii) Straight line from origin; slope = acceleration (g) [1]; $s = ½ \times 10 \times 4 = 20$ m [1].

Q38 **Question:**

[4]

> Element X has 12 protons and 12 neutrons. Element Y has atomic number 17 and mass number 35. Another atom Z has atomic number 17 and mass number 37.

(i) Write the electronic configuration of X. [1]
(ii) How are Y and Z related? [1]
(iii) What ions do X and Y form? Write the formula of the compound formed between X and Y and its formula unit mass (X = 24 u, Y = 35.5 u). [2]

Expected answer / marking points: (i) 2, 8, 2 [1]. (ii) Isotopes (same Z, different mass number) [1]. (iii) X^{2+} and Y^-; XY_2 ($MgCl_2$) [1]; 24 + 71 = 95 u [1].

Q39 **Question:**

[4]

> Night-blooming flowers like Raat ki Rani are white or pale and strongly scented, while many day flowers are bright coloured. Bees, butterflies, moths and bats visit flowers for nectar.

(i) What is pollination? [1]
(ii) Why are night-blooming flowers usually white and fragrant? [1]
(iii) Why is cross-pollination useful for a plant species? What happens after pollination for a seed to form? [2]

Expected answer / marking points: (i) Transfer of pollen from anther to stigma [1]. (ii) White is visible in low light and scent attracts night pollinators like moths/bats [1]. (iii) Brings variation, healthier offspring [1]; pollen tube grows to ovule, male gamete fuses with egg (fertilisation), zygote → embryo, ovule → seed [1].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: